

Chattogram Mohila Polytechnic Institute
Semester Plan, 6th Semester
Computer Science Technology
 Teacher's Name: **Sharmin Akter, Jr. Instructor (Computer)**

Subject code	Subject Name	T P C			Marks				
					Theory		Practical		Total
					TC	TF	PC	PF	
28566	Web Development Project	0	3	1	0	0	25	25	50

8 th	Class-8		Deploy the project • Migrate data from the development environment to the production environment. • Conduct a final round of testing in the production environment. • Ensure all functionalities work as expected
9 th	Class-9		Deploy the project • Deploy the Project to the live environment. • Monitor for any issues during and after deployment. • Provide support
10 th	Class-10	05	Perform the maintenance and monitoring of the project • Implement monitoring tools to track system performance. • Optimize as needed for improved efficiency. • Identify bugs or issues that arise. • Implement updates and enhancements based on user feedback and changing requirements. • Plan for future iterations or releases.
11 th	Class-11	06	Review the project for future plan • Review the project • Identify the Limitation of the project • Prepare further development plans if essential
12 th	Class-12	07	Prepare a project summary • Check the output based on objectives • Demonstrate technical proficiency • Adhere to best practice • Develop team effort • Ready for deployment • Develop learning experience
13 th	Class-13		Project Presentation
14 th	Class-14		Project Presentation

Reference books and websites

1. www.w3schools.com
2. Beginning HTML, XHTML, CSS, and JavaScript - Wrox 1st Edition
3. Beginning PHP, Apache, MySQL Web Development - Wrox 1st Edition

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Class Teacher

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Head of the Department

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Vice-principal

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Principal

Chattogram Mohila Polytechnic Institute

Semester Plan, 4th Semester

Computer Science Technology

Teacher's Name: **Sharmin Akter, Jr. Instructor (Computer)**

Subject code	Subject Name				Marks			
		T	P	C	Theory		Practical	
					TC	TF	PC	PF
28544	Web Design & Development-I	1	6	3	20	30	50	50

Practical

Week	Classes	Experiments Details	
		Number	Name
1st	Class-1	01	INTERPRET WEB TECHNOLOGY AND INDUSTRY REQUIREMENT - Web technology, web development is interpreted. - Front end and back end, full stack developer is stated. - UI/UX design layout is interpreted. - Roles of web developer is stated. - Identify career opportunities
	Class-2		INTERPRET WEB TECHNOLOGY AND INDUSTRY REQUIREMENT - Browser and browser extension are selected and IDE 4 5 (Integrated Development Environment) is ensured. - Task management tools are selected and ensured as per job requirement Task management tools: Jira, Trello, Asana
2 nd	Class-3	02	CONVERT UI/UX TO MARKUP LANGUAGE - Plan a website - Purpose and intended audience of the website are identified. - Design requirements and constraints of using provided templates are identified. - Required design is collected or selected.
	Class-4		CONVERT UI/UX TO MARKUP LANGUAGE - Convert UI/UX to markup language - Given UI/UX design template is converted into required format. - Website layout is developed as per job requirement. - File and folder are named properly and saved in a proper location. - Appropriate markup language is selected. - Markup language: HTML, XML
3rd	Class-5		CONVERT UI/UX TO MARKUP LANGUAGE - Web content is placed into the right position by using markup language tags. - Hyperlinks are added to allow successful navigation between pages of website. - Website is tested to ensure functionality, correct any errors and log in according to testing procedures as per plan.
	Class-6		CONVERT UI/UX TO MARKUP LANGUAGE - Website is opened in a variety of common browsers and check for accessibility, readability, legibility and presentation in accordance with client requirements. - Website is evaluated for fitness purpose in terms of purpose, target audience and specifications of client requirements.
4th	Class-1	03	DEVELOP RESPONSIVE WEBSITE USING FRAMEWORK - Purpose of markup language document is identified. - Accessibility options are identified and determine for visually, physically or otherwise impaired persons. Accessibility: Cultural awareness, Physical impairments, Remote locations
	Class-2		DEVELOP RESPONSIVE WEBSITE USING FRAMEWORK - CSS to markup language documents is linked and verified. - Appropriate styles that are to be controlled by CSS are identified as per responsive design. - Styles are defined and documented in accordance with established design principles.

5th	Class-1		DEVELOP RESPONSIVE WEBSITE USING FRAMEWORK - Create CSS using framework - Frontend framework is identified. Frontend framework: Bootstrap, Tailwind CSS.
	Class-2	04	DEVELOP WEBSITE USING CLIENT-SIDE SCRIPTING LANGUAGE - Plan features - Apply JavaScript - Apply jQuery
6th	Class-1		DEVELOP WEBSITE USING CLIENT-SIDE SCRIPTING LANGUAGE - Apply jQuery - Perform website test
	Class-2		Lab test-1
7th	Class-1	05	DEVELOP WEBSITE USING SERVER-SIDE SCRIPTING LANGUAGE - Plan features - Apply PHP
	Class-2		DEVELOP WEBSITE USING SERVER-SIDE SCRIPTING LANGUAGE Apply PHP
8th	Class-1		DEVELOP WEBSITE USING SERVER-SIDE SCRIPTING LANGUAGE Apply PHP
	Class-2		DEVELOP WEBSITE USING SERVER-SIDE SCRIPTING LANGUAGE Apply PHP
9th	Class-1		DEVELOP WEBSITE USING SERVER-SIDE SCRIPTING LANGUAGE PHP cookie and session are interpreted.
	Class-2		Lab Test-2
10 th	Class-1	06	MANIPULATE DATA USING DATA LAYER Plan data layer
	Class-2		MANIPULATE DATA USING DATA LAYER - Configure embedded ajax and json as required. - Content are added to the site via ajax. - Content are formatted in accordance with client requirements following legislation issues.
11 th	Class-1		MANIPULATE DATA USING DATA LAYER Third party API (Application Programming Interface) data is manipulated by using HTTP (Hypertext Transfer Protocol) methods.
	Class-2		MANIPULATE DATA USING DATA LAYER Third party API: Weather API, Rapid API, Any API, Firebase, AWS API
12 th	Class-1		MANIPULATE DATA USING DATA LAYER Implement database
	Class-2		MANIPULATE DATA USING DATA LAYER CRUD operation is performed (Create, Retrieve, Update, Delete)
13th	Class-1		Perform a final project
	Class-2		Project Presentation
14th	Class-1		Project Presentation
	Class-2		Project Presentation

Reference books and websites

1. www.w3schools.com
2. Learning PHP, MySQL & JavaScript: With jQuery, CSS & HTML5 by Robin Nixon

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Class Teacher

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Head of the Department

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Vice-principal

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Principal

Chattogram Mohila Polytechnic Institute
Semester Plan, 4th Semester
Computer Science Technology
 Teacher's Name: **Sharmin Akter, Jr. Instructor (Computer)**

Subject code	Subject Name				Marks			
		T	P	C	Theory		Practical	
					TC	TF	PC	PF
28544	Web Design & Development-I	1	6	3	20	30	50	50

Theory

Week	Classes	Chapter	
		Number	Name
1st	Class-1	01	WEB TECHNOLOGY AND INDUSTRY REQUIREMENT
2 nd	Class-2		Do
3rd	Class-3	02	UI / UX AND MARKUP LANGUAGE
4th	Class-4		Do
5th	Class-5		Class Test-1
6th	Class-6	03	RESPONSIVE WEBSITE AND FRAMEWORK
7th	Class-7		Do
8th	Class-8	04	CLIENT-SIDE SCRIPTING LANGUAGE
9th	Class-9		Do
10 th	Class-10		Quiz Test-1
11 th	Class-11	05	SERVER-SIDE SCRIPTING LANGUAGE (PHP)
12 th	Class-12		Class Test-2
13th	Class-13	06	DATA MANIPULATION
14th	Class-14		Do
15th	Class-15		Quiz Test-2

Reference books and websites

3. www.w3schools.com
4. Learning PHP, MySQL & JavaScript: With jQuery, CSS & HTML5 by Robin Nixon

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Chattogram Mohila Polytechnic Institute
Semester Plan, 6th Semester
Computer Science Technology
Teacher's Name: **Sharmin Akter, Jr. Instructor (Computer)**

Subject code	Subject Name	T P C			Marks			
					Theory		Practical	
					TC	TF	PC	PF
28561	Database Management System	2	3	3	40	60	25	25
					Total			
					150			

Theory

Week	Class	Chapter	
		Number	Name
1 st	Class-1	01	Basic concept of database system
	Class-2		Do
2 nd	Class-3		Do
	Class-4		Do
3 rd	Class-5	02	Database languages, users, manager and administrator
	Class-6		Do
4 th	Class-7		Do
	Class-8		Class Test-1
5 th	Class-9	03	Data models
	Class-10		Do
6 th	Class-11		Do
	Class-12	04	Relational Database Query Language
7 th	Class-13		Do
	Class-14		Quiz Test-1
8 th	Class-15	05	SQL and PL/SQL
	Class-16		Do
9 th	Class-17		Do
	Class-18		Do
10 th	Class-19	06	Integrity and security
	Class-20		Do
11 th	Class-21	07	Relational database design
	Class-22		Do
12 th	Class-23	08	Data Storage Media
	Class-24		Do
13 th	Class-25		Class Test-2
	Class-26	09	Transactions and concurrency controls
14 th	Class-27		Do
	Class-28	10	Database System Architecture.
15 th	Class-29		Quiz test-2
	Class-30		Student's Problem Solution

REFERENCE BOOKS:

Database System Concepts– Abraham Silberschatz, Henry F. Korth, and S. Sudarshan.

Class Teacher

Head of the Department

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Chattogram Mohila Polytechnic Institute
Semester Plan, 6th Semester
Computer Science Technology
 Teacher's Name: **Sharmin Akter, Jr. Instructor (Computer)**

Subject code	Subject Name	T P C			Marks				
					Theory		Practical		Total
		TC	TF	PC	PF				
28561	Database Management System	2	3	3	40	60	25	25	150

Practical

Week	Classes	Experiment Details	
		Number	Name
1 st	Class-1	01	Install Oracle 19c
2 nd	Class-2	02	Retrieve Data using SQL - Retrieve data from the database using SELECT clause. - Use WHERE to restrict the amount of rows in the result set.
3 rd	Class-3		Retrieve Data using SQL - Use ORDER BY to sort retrieved data. - Use GROUP BY to aggregate data. - Maintain a record of the tasks performed during data retrieval.
4 th	Class-4	03	Use Functions of SQL - Utilize various scalar functions, including LOWER (), UPPER (), INITCAP (). - Apply various aggregate functions, including COUNT (), SUM (), AVG () etc. - Maintain a record of the tasks performed using SQL functions
5 th	Class-5	04	Lab test-1
6 th	Class-6	05	Manipulate Data - Perform Data Manipulation Language (DML) operations on the Database. - Execute the ALTER operation to modify the structure of a table.
7 th	Class-7	06	Manipulate Data 9.3 Maintain a record of the tasks performed during data manipulation.
8 th	Class-8	07	JOIN Tables - Perform Inner Join on tables. - Execute Outer Join on tables.
9 th	Class-9		JOIN Tables - Perform Full Outer Join on tables. - Maintain a record of the tasks performed during table joins
10 th	Class-10	08	Work with Index and Constraints in SQL - Create Index on tables. - Utilize 'DUAL' and SYSDATE for system-related information.
11 th	Class-11		Lab test-2
12 th	Class-12	09	Work with Roles and Privileges - Grant privileges on database users. - Revoke privileges from database users. - Create roles for organized access control.
13 th	Class-13		Work with Roles and Privileges - Assign privileges to roles for efficient management. - Assign roles to users for streamlined access control. - Maintain a record of the tasks performed in managing roles and privileges.
14 th	Class-14		Student's Problem Solution
15 th	Class-15		Student's Problem Solution

REFERENCE BOOKS:

Oracle PL/SQL Programming, Writer Name: Steven Feuerstein, Bill Pribyl, Chip Dawes

Class Teacher

Head of the Department

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Chattogram Mohila Polytechnic Institute
Semester Plan, 6th Semester
Computer Science Technology
 Teacher's Name: Sharmin Akter, Jr. Instructor (Computer)

Subject code	Subject Name	T P C			Marks				
					Theory		Practical		Total
					TC	TF	PC	PF	
28565	Surveillance Security System	1	3	2	20	30	25	25	100

Practical

Week	Classes	Experiment Details	
		Number	Name
1st	Class-1	01	Requirements, Quotation and approval. - Collect requirements from Client. - Select products as per requirement. - Prepare Budget as per requirements. - Prepare design and diagram - Prepare a quotation as per requirements. - Approve quotation from client. - Maintain the record of performed task
2 nd	Class-2	02	Testing equipment's and set up. - Collect the necessary equipment's and select BNC or Video balun, Power Adapter, Cable, Camera, DVR and NVR. - Connect and set up equipment's according to diagram. - Test the equipment's. - Identify and Isolate the faulty device. - Maintain the record of performed task.
3rd	Class-3	03	Perform Wiring of Power and Network Cable. - Follow Occupational safety and health (OSH) practices. - Identify the power source. - Perform wiring as per required. - Install power equipment. - Connect cable according to diagram. - Connect different network equipment. - Maintain the record of performed task.
4th	Class-4	04	Install the CCTV camera. - Make survey and identify the location of the camera to be fixed. - Select the suitable camera depending on the coverage area required by the customer. - Connect power equipment of different CCTV Camera. - Select cable and connector (coaxial /twisted pair cable and BNC connectors / Video Balun). - Connect all the cables and cameras as per procedure. - Ensure all cable and connectors are connected properly. - After complete all procedure connect AC/DC Power Adapter with CCTV Camera.
5 th	Class-5		Repeat number 4 experiment

6 th	Class-6	O5	Install the IP (and PTZ) camera - Follow installation procedures as per manuals. - Mount the IP camera s such as pan, tilt, zoom. - Connect power equipment / Use POE switch of IP Camera s. - Assign IP address. - Maintain the record of performed task.
7 th	Class-7	06	Install and configure DVR/NVR Machine. - Set up DVR/NVR as per diagram. - Install Hard disk in DVR/NVR. - Ensure that all cables and connectors are connect Properly. - Assign IP Address to connect DVR/NVR. - Install the appropriate software for remote monitoring. - Configure the DVR/NVR as per procedure. - Maintain the record of performed task.
8 th	Class-8		Repeat number 6 experiment
9 th	Class-9		Lab Test-1
10 th	Class-10	05	Configure the camera - Perform control and configure/settings of camera such rotation, speed of movement in pan/tilt camera - Use stable mounting structure and ensure that is not disturbance any circumstance. - Ensure that cameras are protected from light while installing in outdoor. - Maintain the record of performed task
11 th	Class-11	06	Backup data from DVR/NVR - Connect external storage device to the DVR/NVR. - Use Backup Option from the DVR/NVR as per manual. - Select particular camera for backup the footage. - Set time and date to find the footage for backup. - select video format for backup such as dav, mp4, PAL, NTSC. - Start or Click Backup option. - Maintain the record of performed task.
12 th	Class-12		Repeat number 5 & 6 experiments.
13 th	Class-13		Lab Test-2
14 th	Class-14		Student's Problem Solution

Reference books

- Digital Video Surveillance and Security - by Anthony C. Caputo
- Digital CCTV -by Emily Harwood

Class Teacher

Head of the Department

Vice-principal

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Chattogram Mohila Polytechnic Institute
Semester Plan, 6th Semester
Computer Science Technology
 Teacher's Name: Sharmin Akter, Jr. Instructor (Computer)

Subject code	Subject Name	T P C			Marks				
					Theory		Practical		Total
		TC	TF	PC	PF				
28565	Surveillance Security System	1	3	2	20	30	25	25	100

Theory

Week	Classes	Experiment Details	
		Number	Name
1st	Class-1	01	Surveillance security System - Define surveillance system. - State different types of surveillance.
2 nd	Class-2		Surveillance security System - Describe problems & constrains of surveillance. - State the uses of access control in Surveillance systems.
3rd	Class-3	02	Video surveillance - Define video surveillance. - Illustrate the Construction procedure of a video surveillance system. - State the function of video surveillance. - Explain the essential components of the CCTV camera system
4th	Class-4		Video surveillance - Explain construction procedure of various nodes and blocks in CCTV surveillance system - Explain Security policy model.
5 th	Class-5		Class Test-1
6 th	Class-6	O3	Cameras, Lens and sensors - Define pixels, camera, Lens and Sensor. - Describe the function of different types of cameras.
7 th	Class-7		Cameras, Lens and sensors - List different types of lenses - State Fixed lenses, zoom lenses, Varifocal lenses, Monofocal lenses, Wide Angle lenses with uses - State various types of CCTV drawing Symbols.
8 th	Class-8		Quiz Test-1
9 th	Class-9	04	DVR, NVR interface and Remote access - Define DVR and NVR. - Describe the function of various blocks of DVR and NVR. - State the uses of physical storage and cloud storage.
10 th	Class-10		DVR, NVR interface and Remote access - Describe the recording format of a DVR and NVR. - Define remote access system. - Mention the Advantage and disadvantage of remote access system
11 th	Class-11		Class Test-2
12 th	Class-12	05	Surveillance system installation

			- List the necessary equipment of Surveillance system. - State the various type of cables uses in surveillance system
13 th	Class-13		Surveillance system installation - Describe the function and operation of Biometric device, Intercom, Door lock, Emergency Alarm - Illustrate the installation process of Biometric system. - Describe the maintenance procedure of Surveillance system
14th	Class-14		Quiz Test-2

Reference books

- Digital Video Surveillance and Security - by Anthony C. Caputo
- Digital CCTV -by Emily Harwood

Class Teacher

Head of the Department

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Chattogram Mohila Polytechnic Institute
Semester Plan, 2nd Semester
Computer Science Technology
 Teacher's Name: Sharmin Akter, Jr. Instructor (Computer)

Subject code	Subject Name	T P C	Marks					
			Theory		Practical		Total	
			TC	TF	PC	PF		
28522	Computer Graphics Design-1	0	6	2	0	0	50	50
								100

Practical

Week	Classes	Experiment Details	
		Number	Name
			Separate and compose Images
1 st	Class-1	01	Identify image source
	Class-2	02	Identify image standards
2 nd	Class-3	03	Separate Images using magic wand
	Class-4	04	Separate Images using lasso tools
3 rd	Class-5	05	Separate Images using pen tools
	Class-6	06	Create layer and compose
4 th	Class-7	07	Evaluate own work
	Class-8		Lab Test-1
5 th			Create basic designs using illustration software
	Class-9	08	Create basic designs
6 th	Class-10	09	Create Outline and transfer.
	Class-11	10	Develop conceptual skills and ideas
7 th	Class-12	11	DO
	Class-13	12	DO
8 th	Class-14	13	DO
	Class-15	14	Evaluate own work
			Manipulate image using image processing Software
9 th	Class-16	15	Retouch Image
	Class-17	16	Colour Correction
10 th	Class-18	17	Apply Effect
	Class-19	18	Evaluate own work
11 th	Class-20	19	Lab Test-2
			Create professional designs using Illustration software
	Class-21	20	Prepare design
12 th	Class-22	21	Create Design
	Class-23	22	Review and Finalize
13 th	Class-24	23	Lab Test-3
	Class-25		Student Problem Solution
14 th	Class-27		Student Problem Solution
	Class-28		Student Problem Solution

Class Teacher

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